



EASY VISA

Machine Learning 2



FEBRUARY 9, 2025

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CONTEXT

The United States faces a high demand for skilled human resources, with companies actively seeking qualified professionals both domestically and internationally. However, one of the primary challenges remains identifying and attracting the right talent while ensuring compliance with labor laws.

The **Immigration and Nationality Act (INA)** allows foreign workers to enter the U.S. on a temporary or permanent basis while protecting domestic workers' rights. The **Office of Foreign Labor Certification (OFLC)** oversees the certification process for employers who seek to hire foreign workers. Employers must demonstrate that there are not enough U.S. workers available at competitive wages before hiring internationally.

As the demand for foreign labor increases, the volume of visa applications has risen significantly, making manual review processes **time-consuming and inefficient**. The need for a data-driven solution is essential to streamline visa approvals while maintaining compliance with labor regulations.

Objective

In **FY 2016**, the OFLC processed **775,979 employer applications** for over **1.69 million positions**, marking a **9% increase from the previous year**. With the continuous rise in applications, manually reviewing each case has become increasingly challenging.

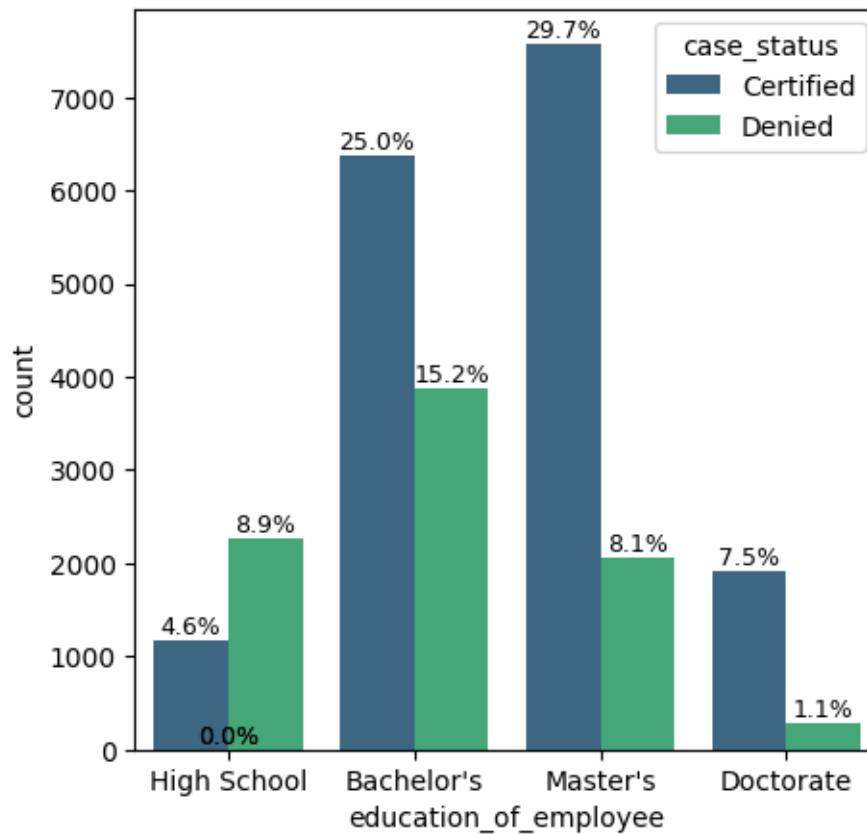
To address this issue, the OFLC has engaged **EasyVisa**, a data-driven consulting firm, to develop a **Machine Learning-based solution** that can:

- **Automate the visa approval screening process** by identifying candidates with higher chances of approval.
- **Analyze key factors** influencing visa approvals, such as education level, job experience, and prevailing wage.
- **Recommend a suitable applicant profile** based on the factors that significantly impact case status (certified or denied).

By leveraging machine learning, this solution aims to optimize the visa application process, reduce manual workload, and ensure fair and efficient decision-making.

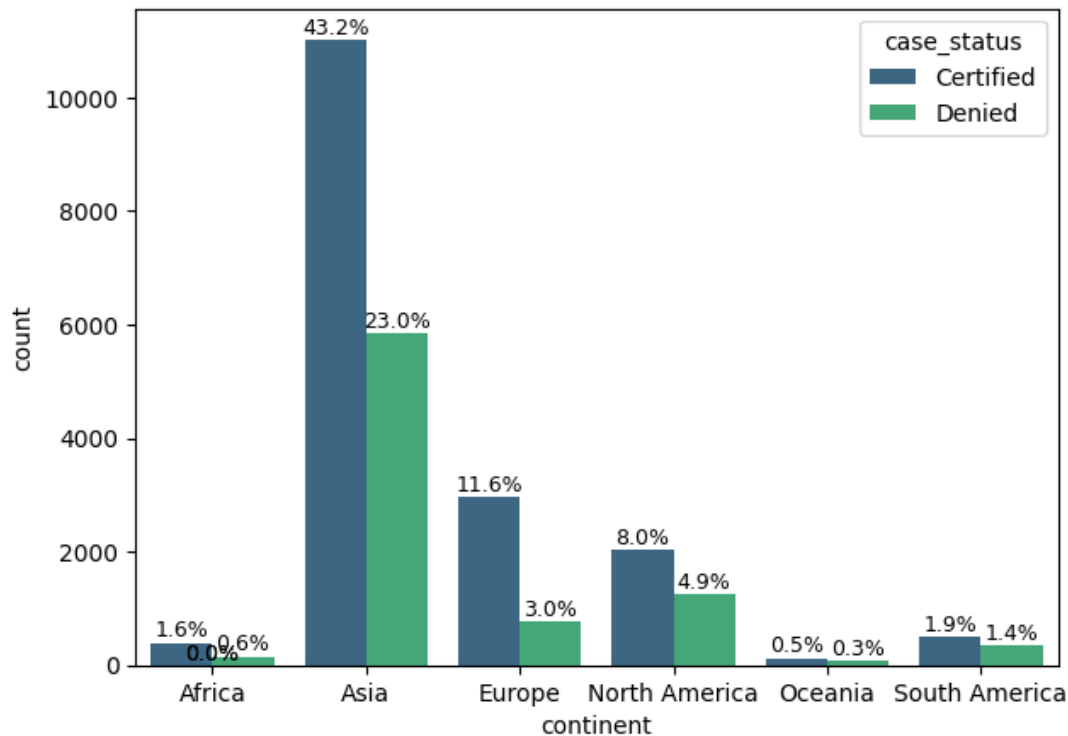
Exploratory Data Analysis (EDA)

1. Those with higher education may want to travel abroad for a well-paid job. Does education play a role in Visa certification?



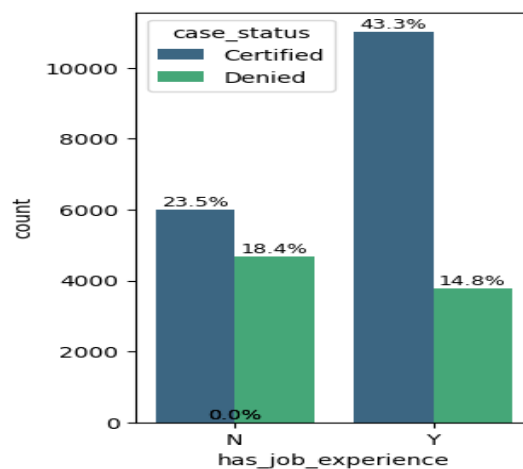
- As the graph above shows, the ratio of applications being certified versus denied increases considerably as an applicant's highest level of education achieved increases.
- The ratio of an applicant with a high school diploma being approved versus denied is ~1:2, whereas the same ratio for an applicant with a doctorate is ~7:1.

2. How does the visa status vary across different continents?



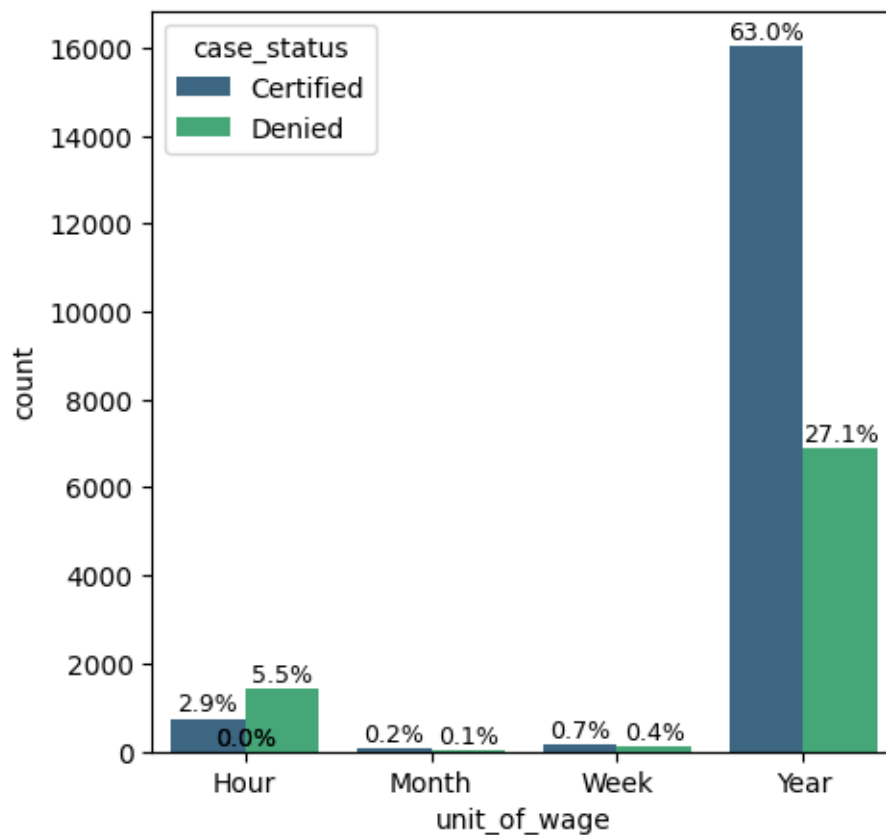
- Applicants from Asia comprise ~2/3 of all applications and these applicants have almost a 2:1 ratio of approvals to denials.
- An application from a European applicant has the best ratio of approvals to denials (~4:1).

3. Experienced professionals might look abroad for opportunities to improve their lifestyles and career development. Does work experience influence visa status?



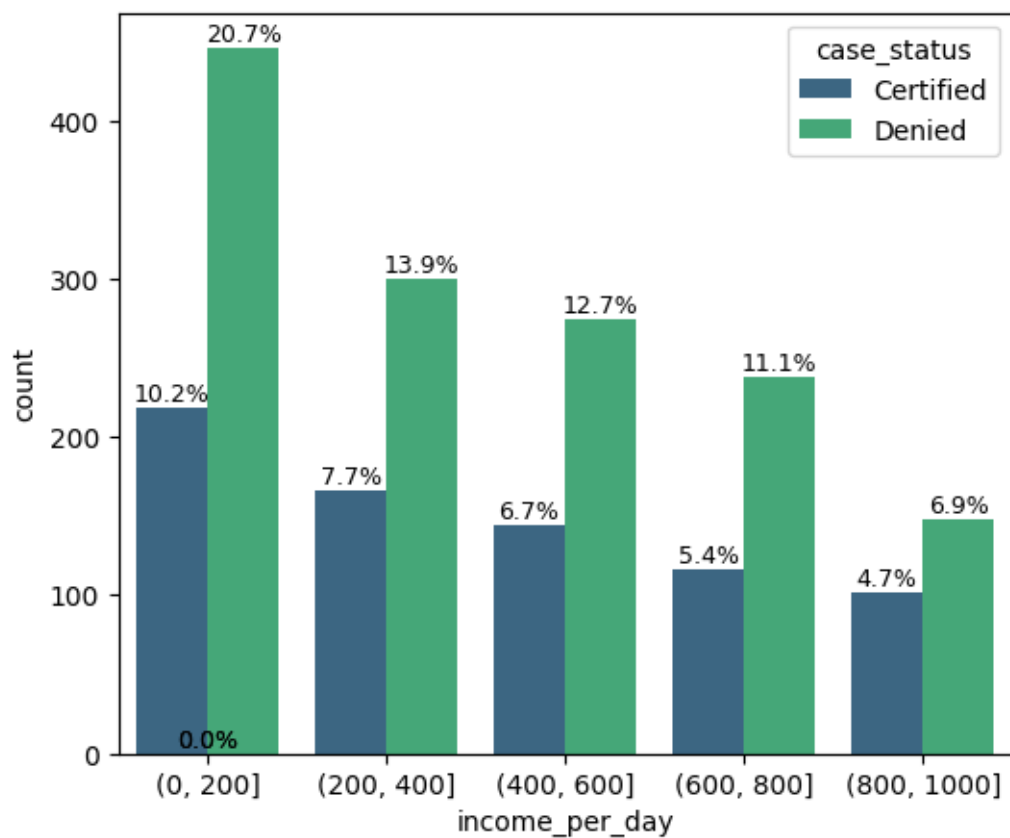
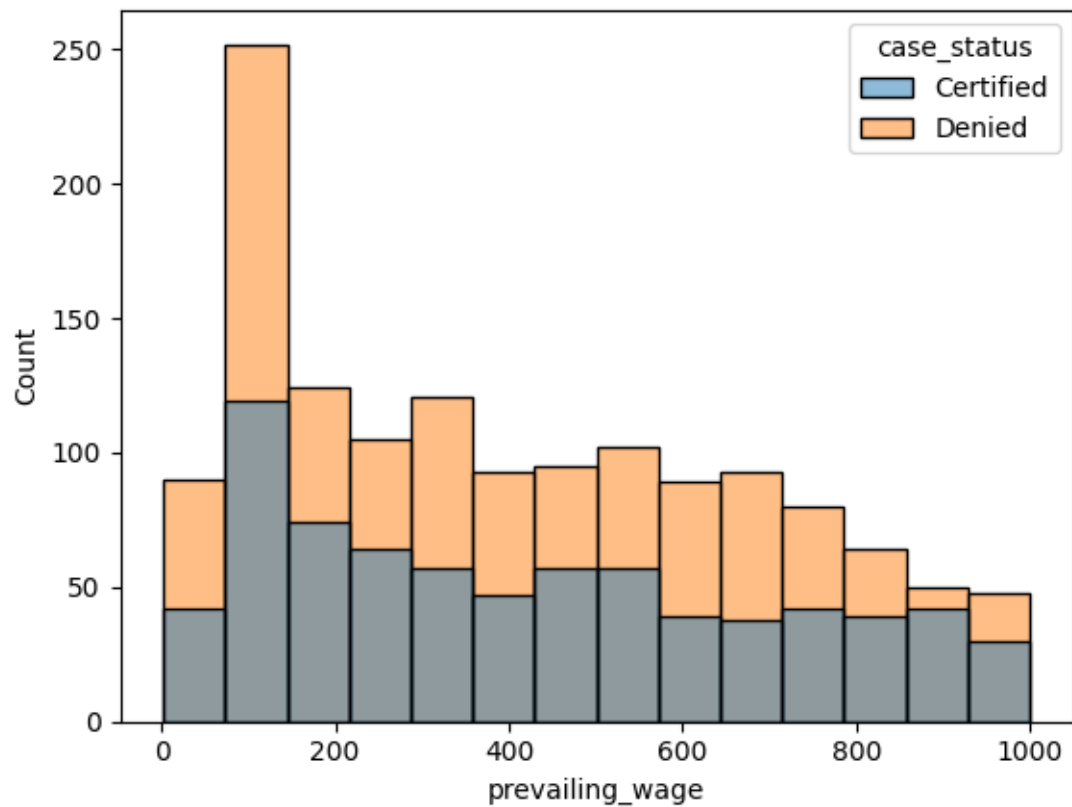
- Applicants with job experience have a ratio of approved to denied applications of ~3:1, whereas the same ratio for applicants without job experience have around a 5:4 ratio (i.e., approximately equivalent)

4. In the United States, employees are paid at different intervals. Which pay unit is most likely to be certified for a visa?



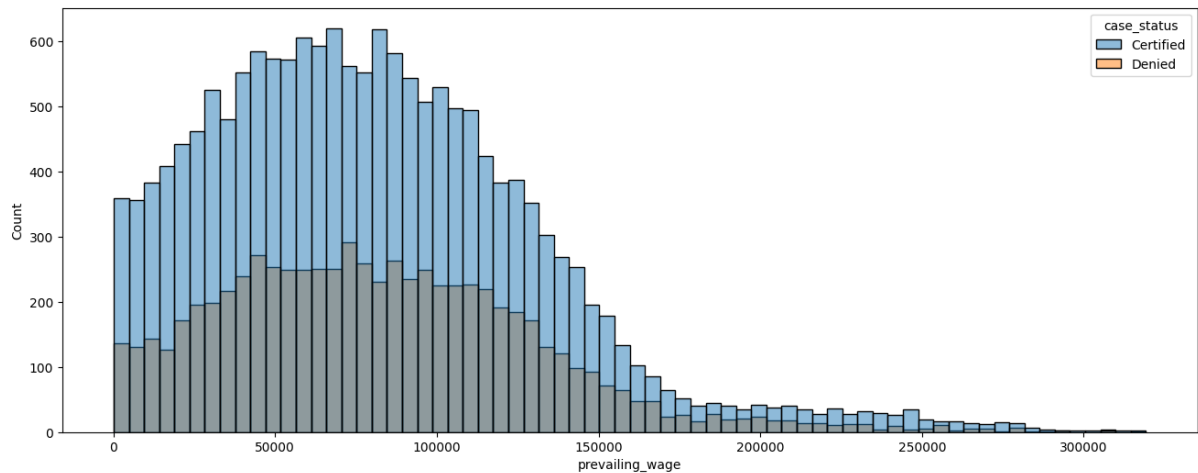
- Applicants who are applying to work in a job with an hourly rate have a ratio of approved versus denied applications of ~1:2. Additionally, these applicants comprise only ~8% of all applications, but comprise ~17% of all denials.
- Applicants from any other *unit_of_wage* category have a ratio of ~2:1, with applications for jobs with annual salaries showing nearly a 2.5:1 ratio of approvals to denials.

5. The US government has established a prevailing wage to protect local talent and foreign workers. How does the visa status change with the prevailing wage?

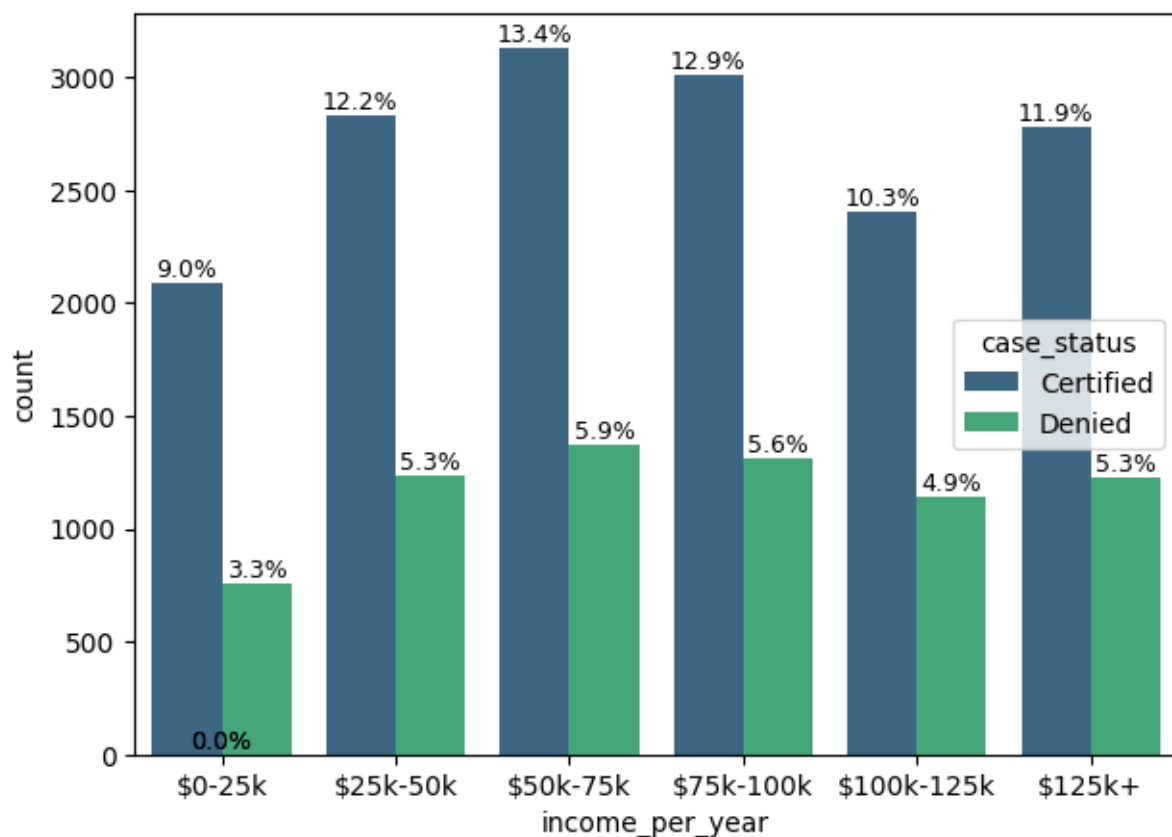


- As the graphs above show, even within applicants for jobs with an 'Hour' *prevailing_wage*, the ratio of approved to denied applications increases considerably as *prevailing_wage* increases.

- Even at the highest *income_per_day* category, an application is still more likely to be denied than approved.

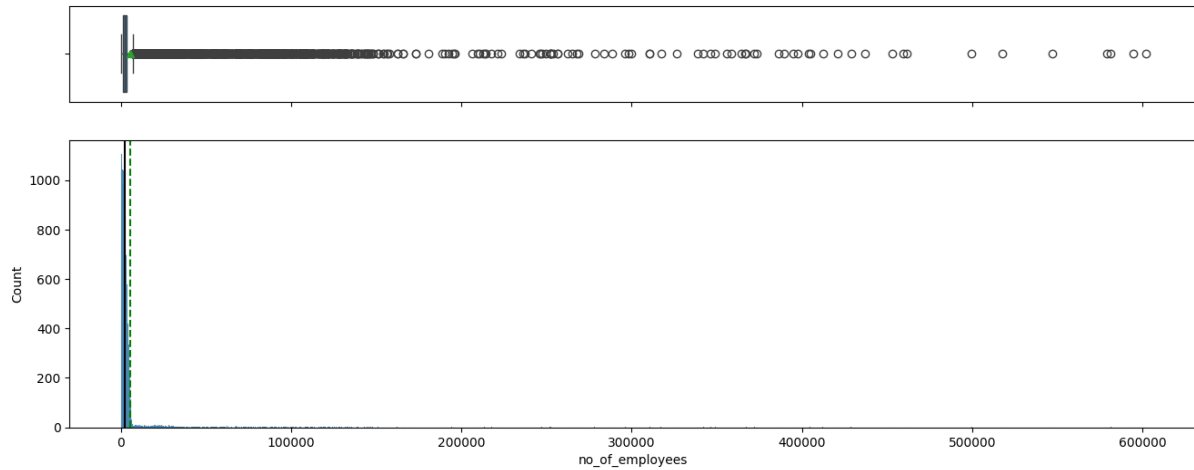


- The distribution of wages is (as expected) right skewed, indicating a long-tail of jobs with annual salaries above 150,000 dollars.
- As the distribution above shows, the ratio of approved to denied applications remains approximately equivalent across the range of wages.

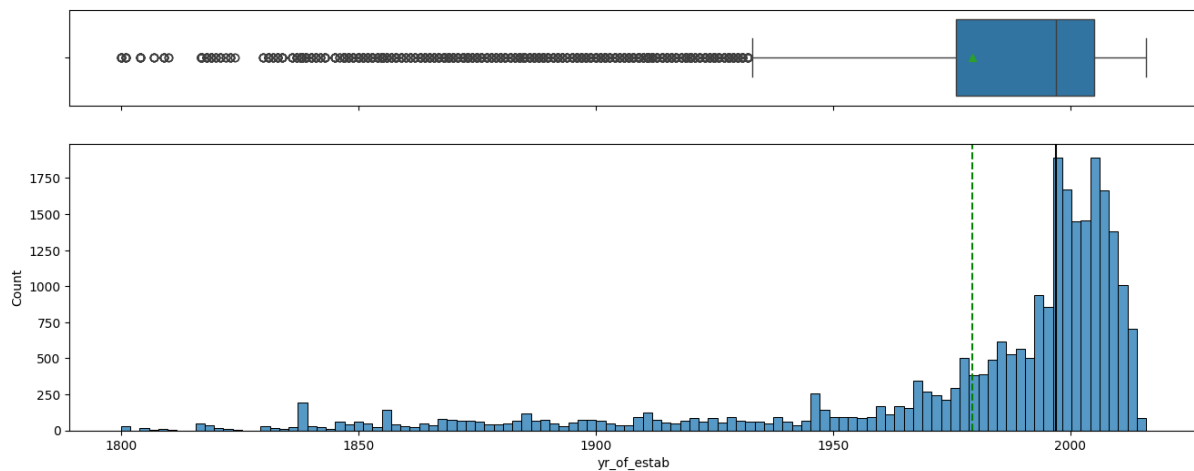


- In support of the observation above, there does not appear to be a significant change in ratio of approved to denied applications as *income_per_year* increases. To the contrary, the ratio may actually decrease as this *prevailing_wage* increases.

- For applications for jobs with an *income_per_year* wage, the highest concentration of applications is for jobs with 50,000 to 75,000 dollars per year in salary.

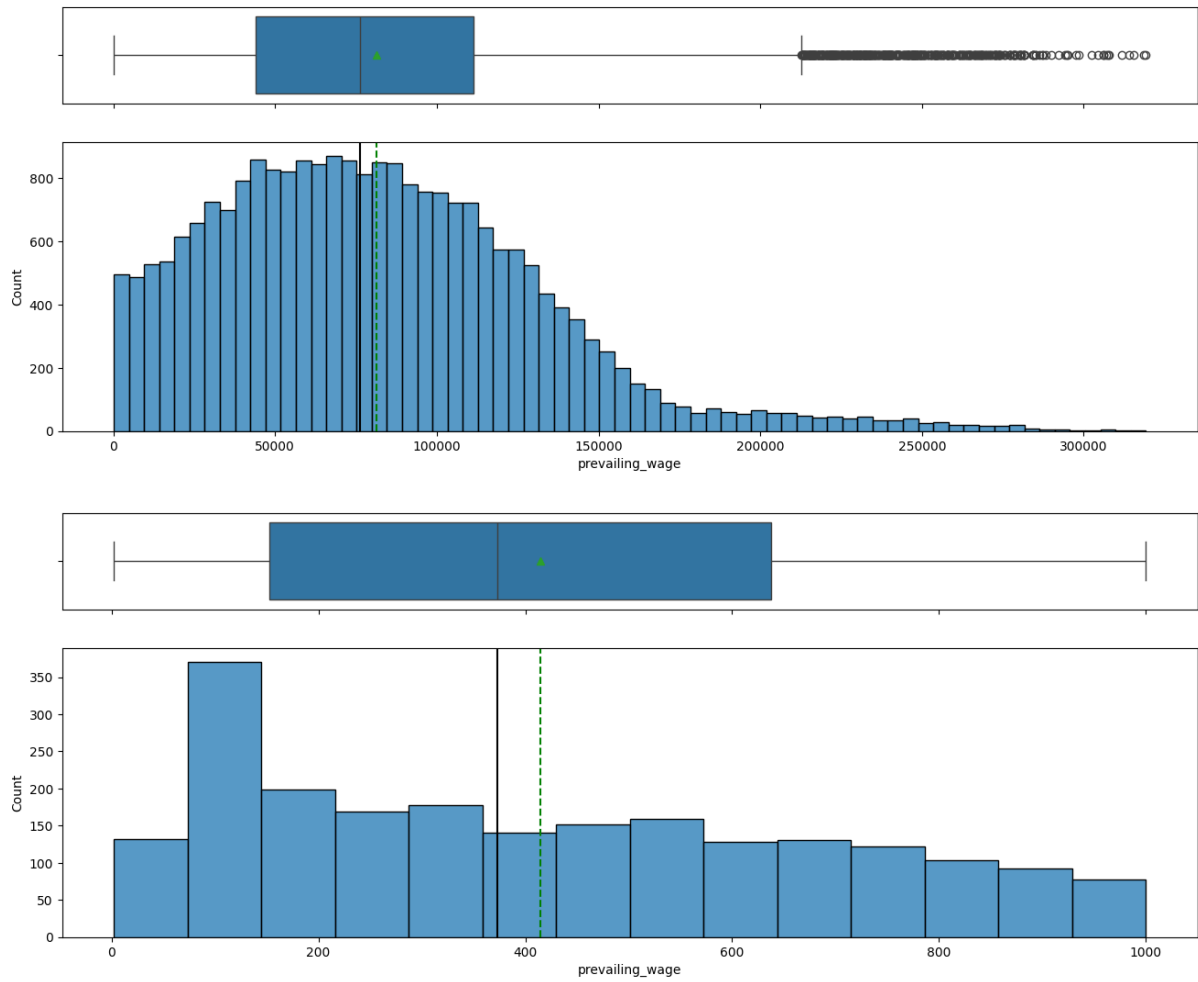


- The data and graphs above, indicate the distribution of companies by number of employee is heavily right skewed, but without any outlier values that require further investigation or treatment.

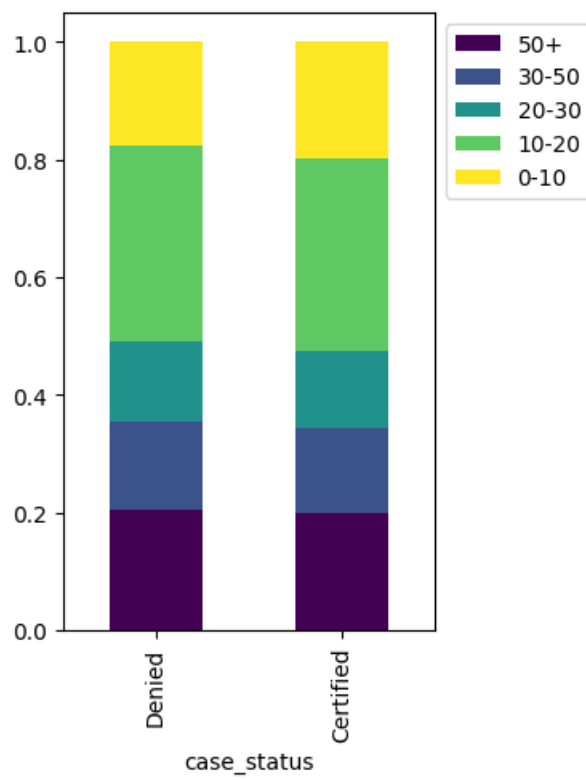
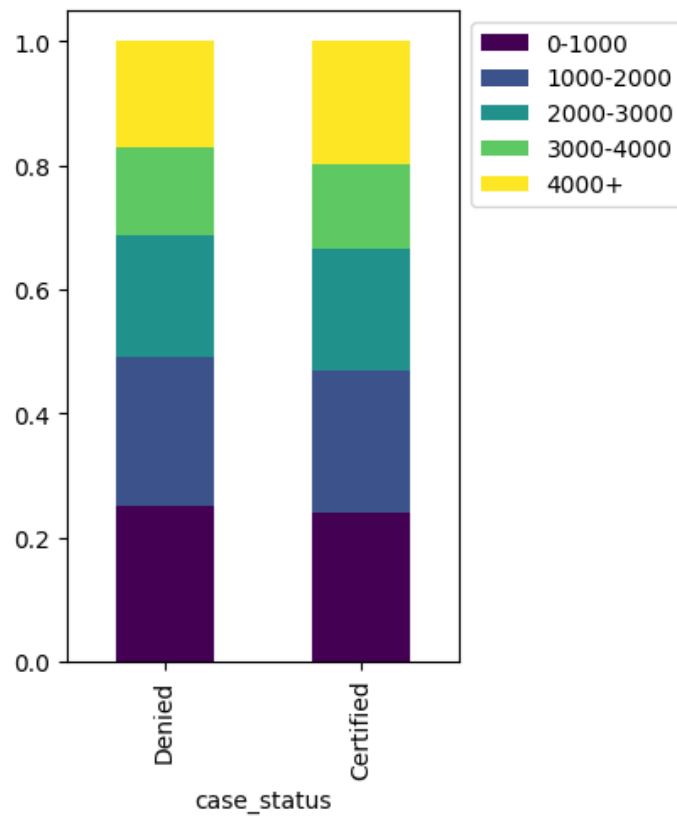


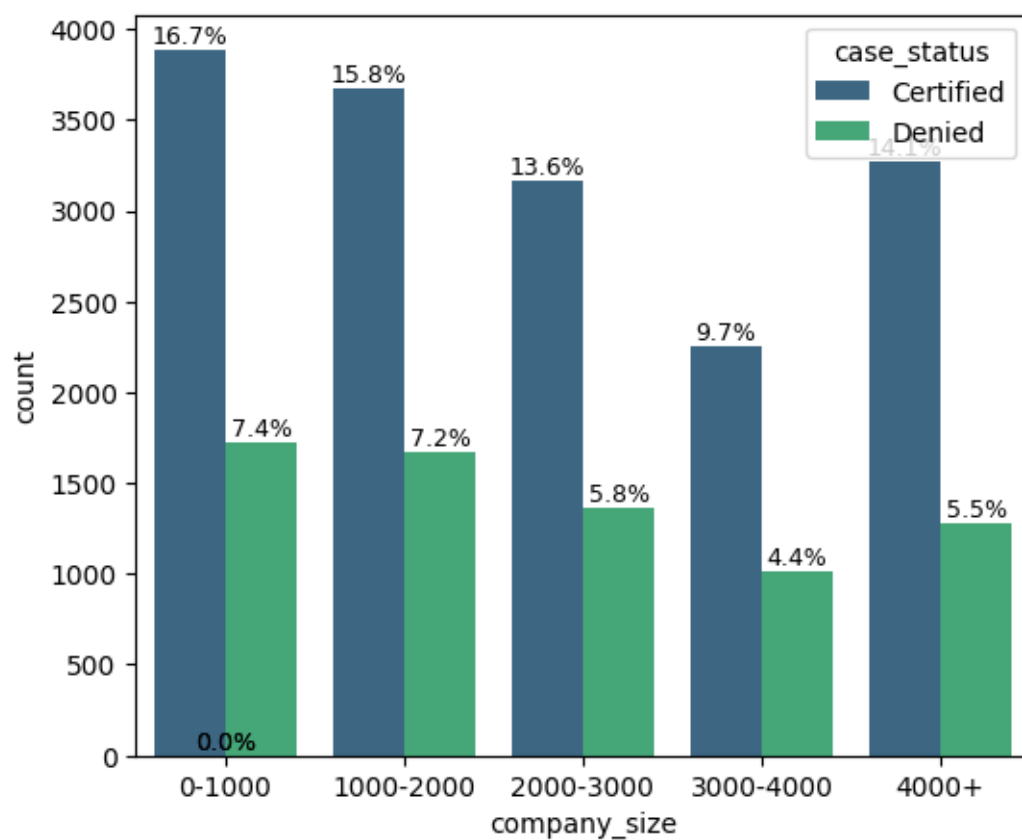
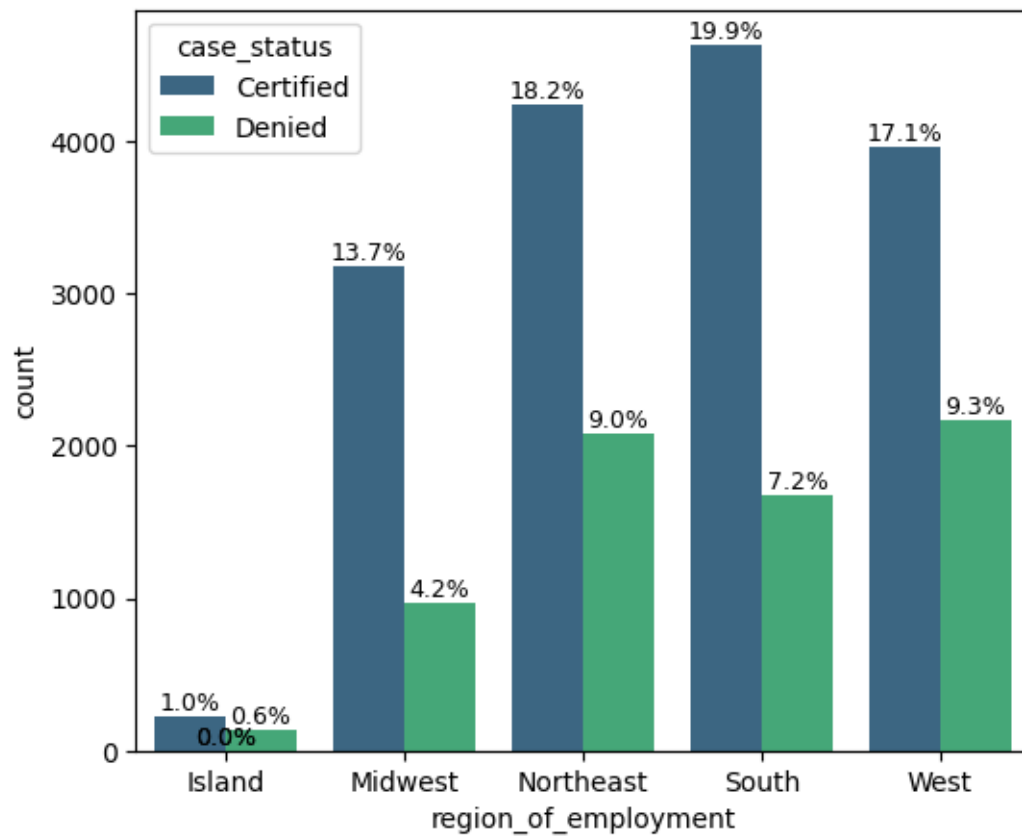
- The data and graphs above, indicate the distribution of companies by year of establishment (i.e., founding) is heavily left skewed, but without any outlier values which require further investigation or treatment.

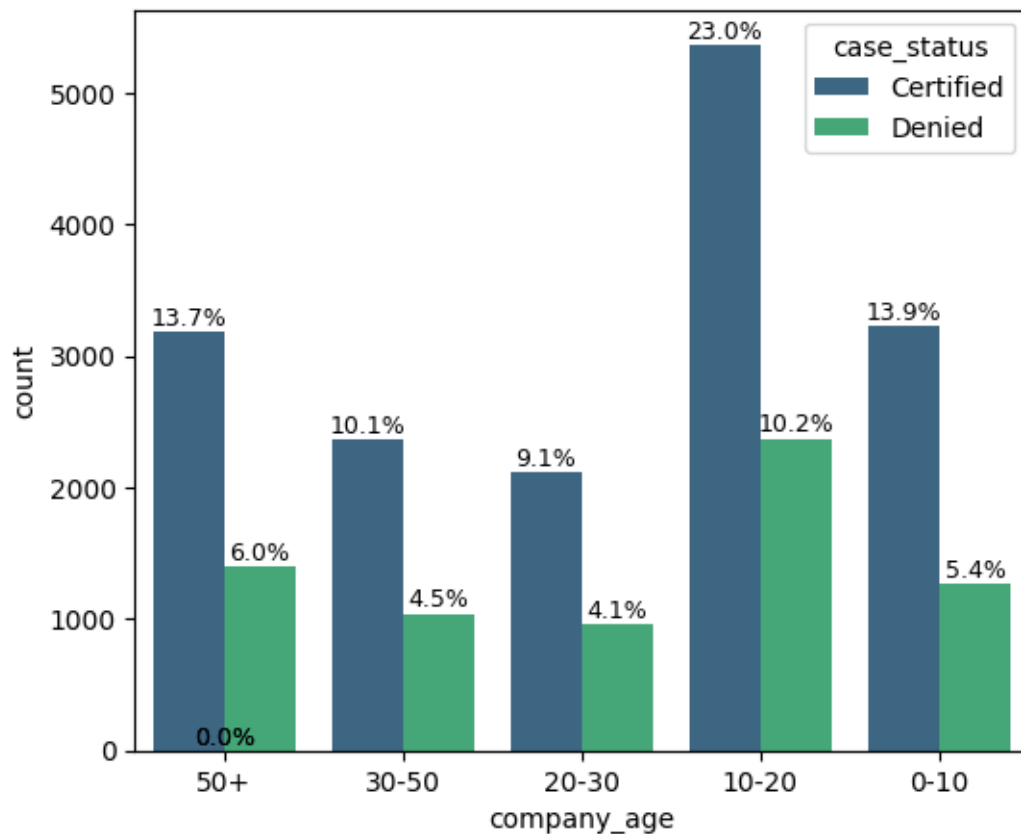
Summary of float64 variables



Bivariate Analysis







Additional EDA observations

- In addition to the analysis above on the ratio of approved to denied applications by education, previous job experience, continent, prevailing wage, and unit of wage:
 - If a job does or does not require training does not affect this ratio.
 - If a job is a full-time position or not does not affect this ratio.
 - If a job is located in the Midwest or South region, applicants have a higher ratio of approvals than those in the Northeast, West, and particularly the Island region.
 - Jobs that applicants with doctorate degree are applying for jobs have the lowest mean prevailing wage of any education.
 - Temporary (i.e., not full-time) jobs that applicants are applying for have a significantly higher prevailing wage than full-time positions.

Bagging and Boosting models

Model evaluation criterion

Model can make wrong predictions as:

1. Predicting an applicant should be approved the applicant is denied.
2. Predicting an applicant should be denied and the applicant is approved.

Which case is more important?

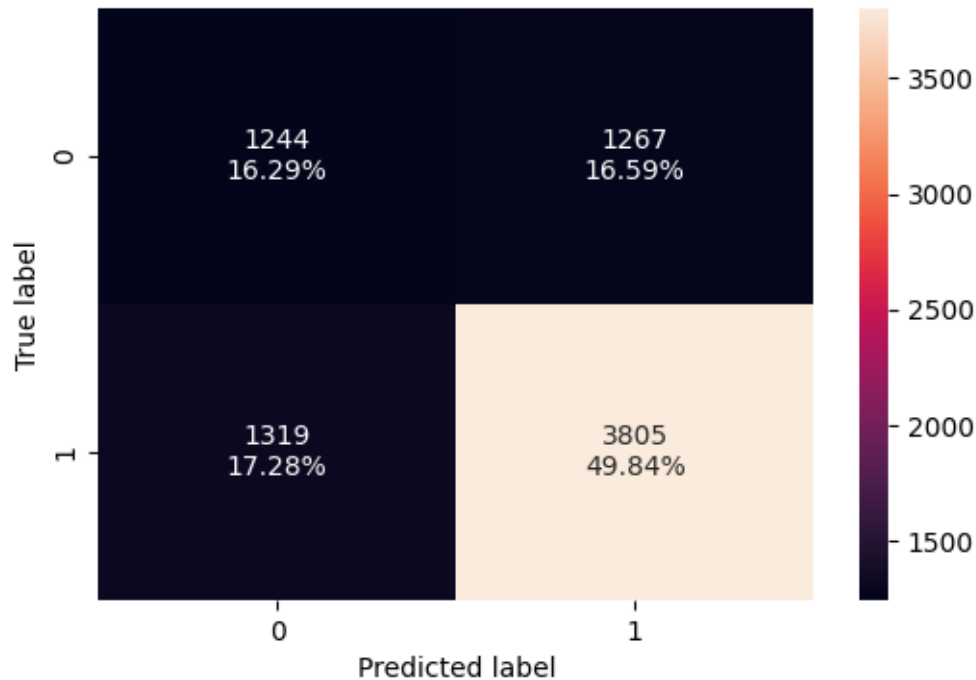
- Both are important:
 - If an applicant is approved when they would have been denied, an unqualified employee will get a job that should have been filled by a US citizen.
 - If an applicant is denied when they should have been approved, U.S. companies will not be able to fill critical positions and the overall economy will not be as productive.

How to reduce the losses?

- As the process of reviewing each application is time and resource-intensive, this model should identify those candidates predicted to be approved, so agents can prioritize these applications.
- F1 Score can be used as the metric for evaluation of the model, as the greater the F1 score, the higher the chances of minimizing False Negatives and False Positives.
- We will use balanced class weights, where applicable, so that model focuses equally on both classes.

Decision Tree Model

We will build our model using the DecisionTreeClassifier function, using default 'gini' criteria to split.

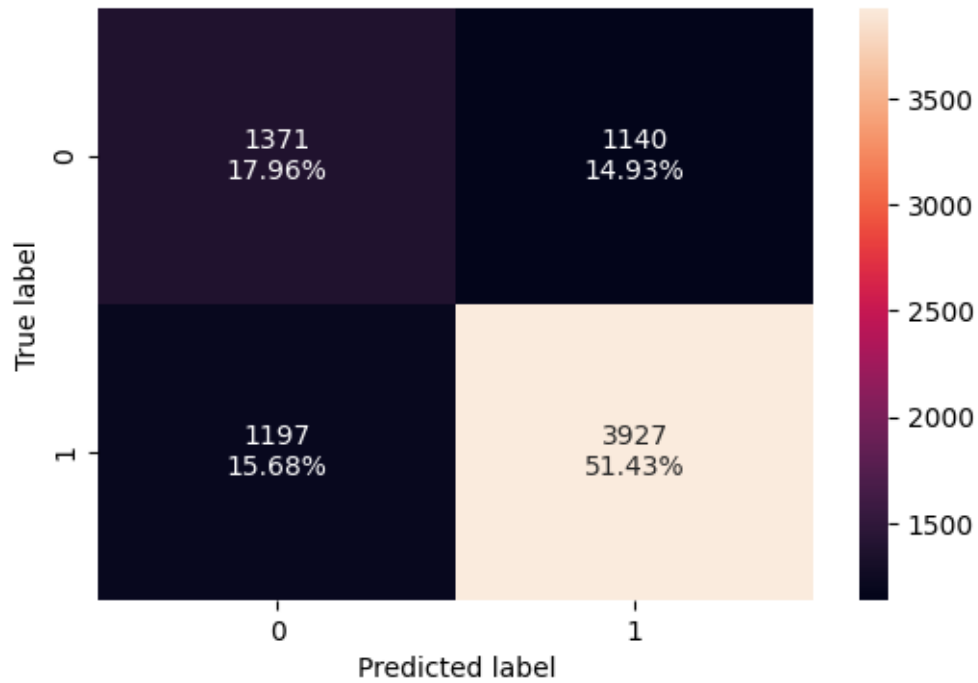


Confusion Matrix -

- Applicant was approved and the model predicted approval : True Positive (observed=1,predicted=1)
- Applicant was denied and the model predicted approval : False Positive (observed=0,predicted=1)
- Applicant was denied and the model predicted denial : True Negative (observed=0,predicted=0)
- Applicant was approved and the model predicted denial : False Negative (observed=1,predicted=0)

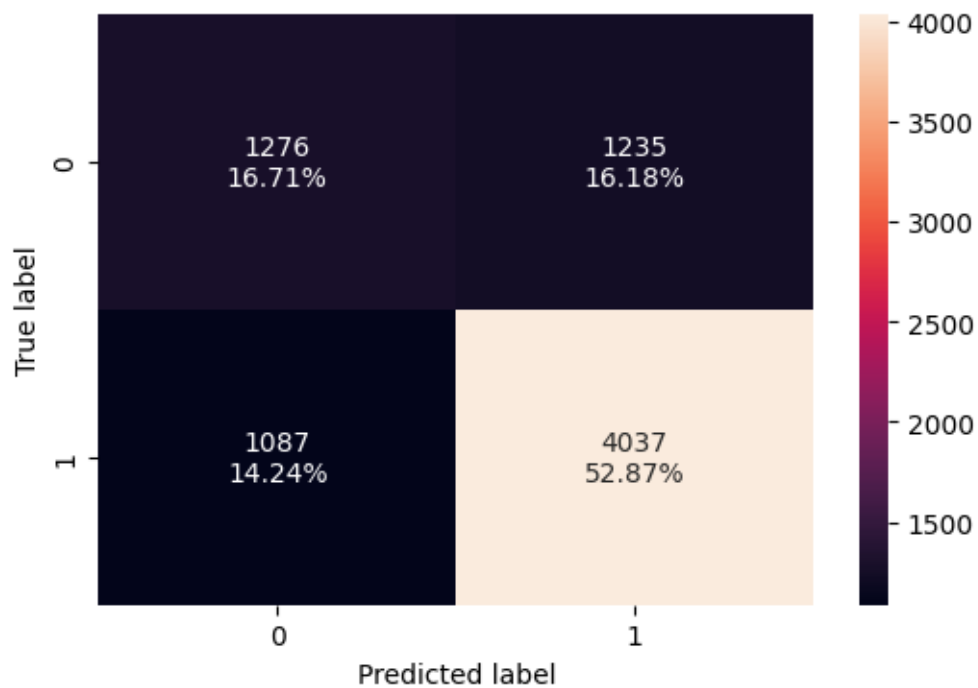
The Decision Tree is working well on the training data, but is not able to generalize well on the test data, indicating the model is overfit.

Bagging Classifier



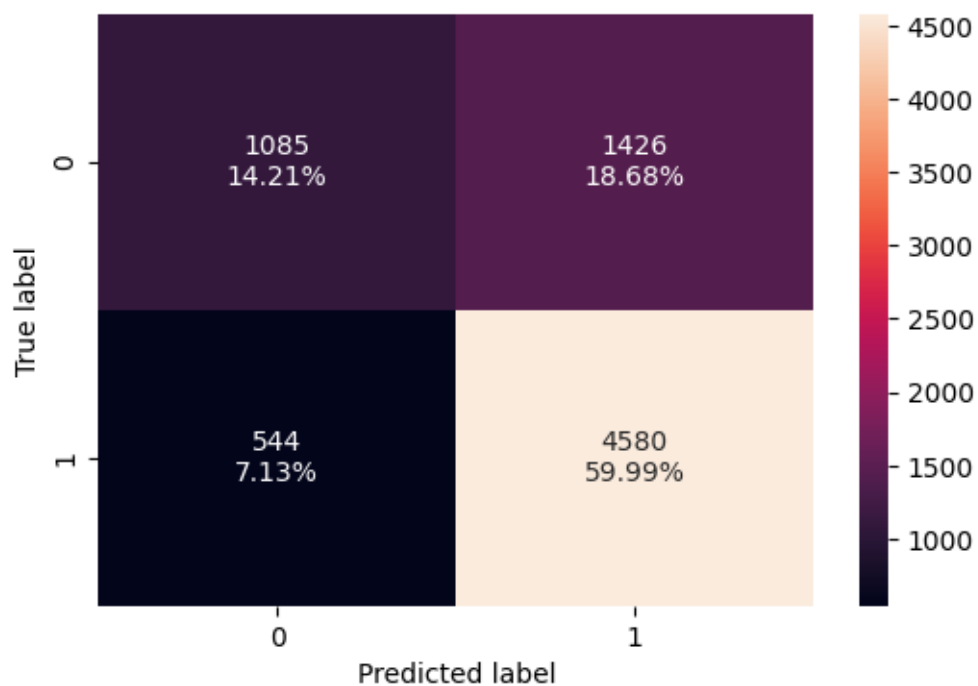
- Similar to the Decision Tree model, this bagging classifier is overfit to the training dataset but has a higher F1 score.

Random Forest



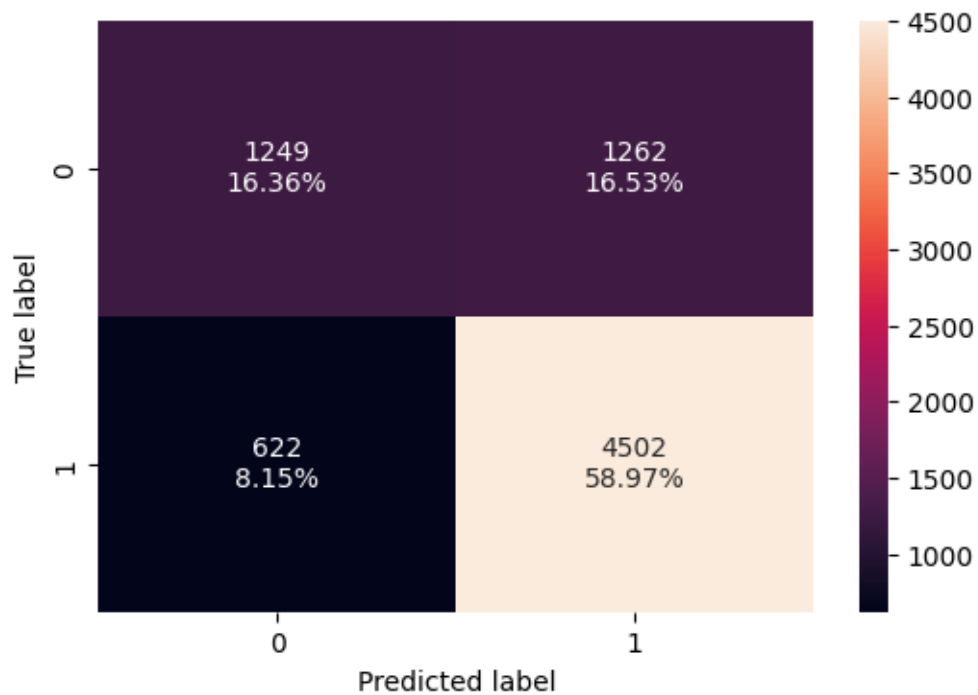
- Similar to the models above, this Random Forest is also overfit to the training dataset but has the highest F1 score so far.

AdaBoost Classifier



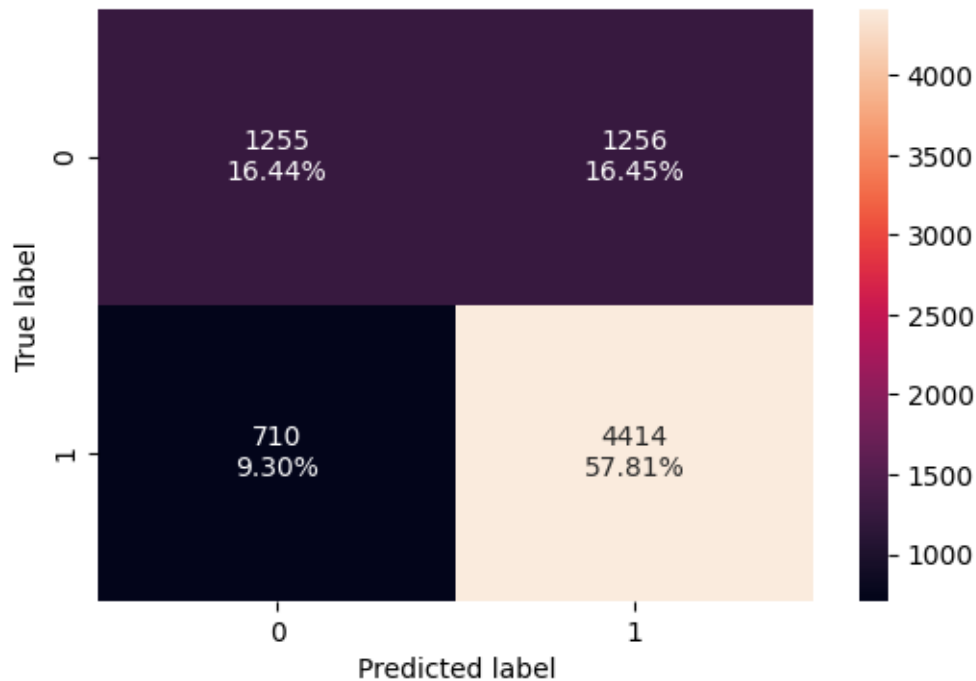
- This AdaBoost classifier performs equally well on the training and testing datasets, indicating that the model is not overfit.
- Additionally, the model performs the best of any model yet (i.e., highest F1 score) on the testing data.

Gradient Boosting Classifier



- This Gradient Boosting classifier performs equally well on the training and testing datasets, indicating that the model is not overfit.
- Additionally, this model performs even better than the AdaBoost model above.

XGBoost Classifier



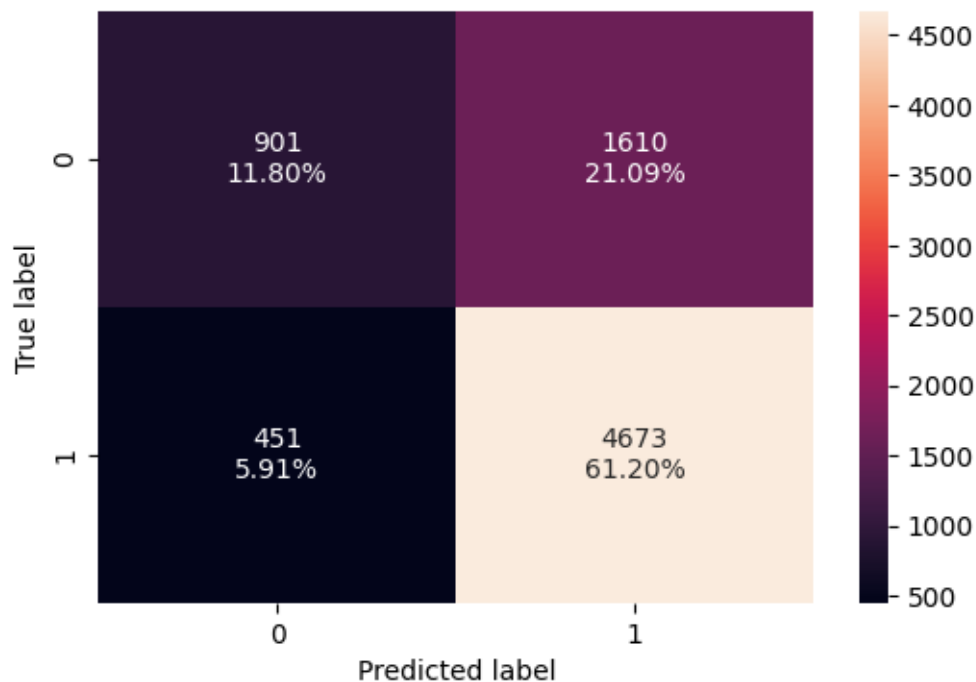
- This XGBoost classifier performs equally well on the training and testing datasets, indicating that the model is not overfit.
- Additionally, this model performs as well as the AdaBoost model and slightly worse than the Gradient Boost model on the test data.

Overall, the bagging models perform best on the training data, due to all three of the models overfitting this data set.

As shown above, all three of the boosting models do not overfit the data and achieve higher F1 scores on the training data set, with the Gradient Boosting classifier performing the best.

Hyperparameter tuning

Tuning Decision Tree



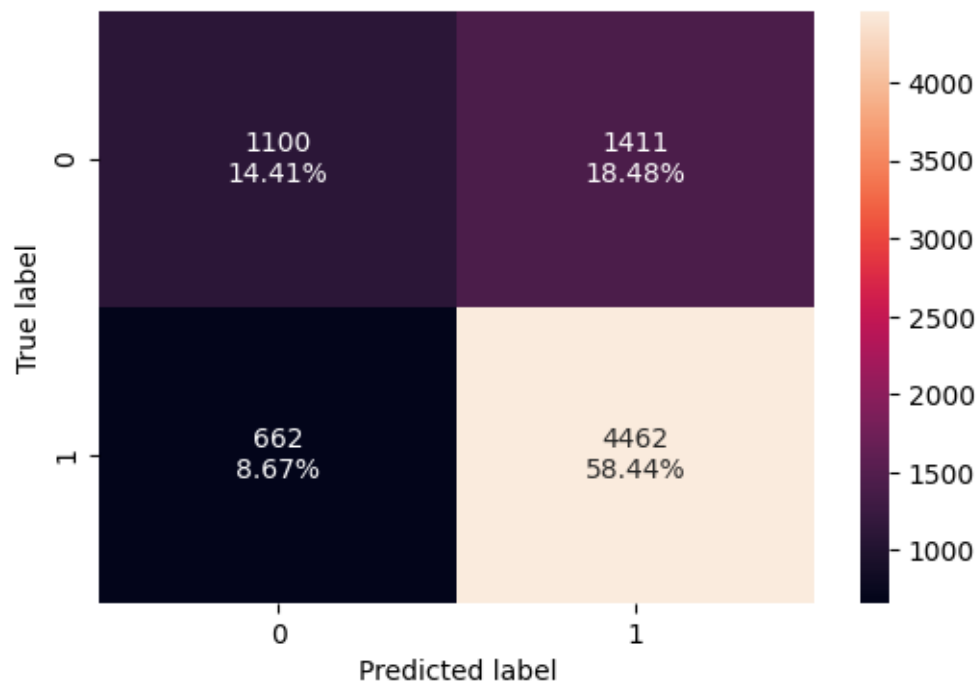
- The Decision Tree model after tuning performs equally well on the training and testing datasets, indicating that the model is not overfit.
- Additionally, this model performs as well as two out of the three boosting models above on the test data.
- Both of these indicate this tuned model is a significant improvements over the base model.

Tuning Bagging Classifier

Hyperparameters available for bagging classifier include:

- `base_estimator`: The base estimator to fit on random subsets of the dataset. If `None`(default), then the base estimator is a decision tree.
- `n_estimators`: The number of trees in the forest, default = 100.
- `max_features`: The number of features to consider when looking for the best split.
- `bootstrap`: Whether bootstrap samples are used when building trees. If `False`, the entire dataset is used to build each tree, default=`True`.
- `bootstrap_features`: If it is true, then features are drawn with replacement. Default value is `False`.

- `max_samples`: If `bootstrap` is `True`, then the number of samples to draw from `X` to train each base estimator. If `None` (default), then draw `N` samples, where `N` is the number of observations in the train data.
- `oob_score`: Whether to use out-of-bag samples to estimate the generalization accuracy, `default=False`.

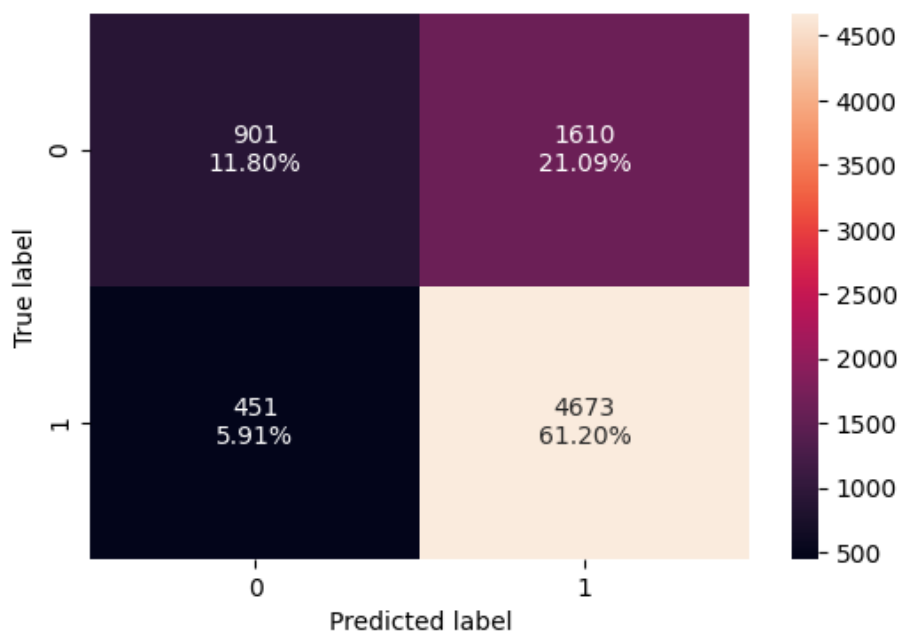


- The Bagging Classifier after tuning performs far less well on the test data than on the training data, indicating that the model is still overfit.
- However, this model does perform better than the base model.

Tuning Random Forest

Hyperparameters for Random Forest include:

- `n_estimators`: The number of trees in the forest, default = 100.
- `max_features`: The number of features to consider when looking for the best split.
- `class_weight`: Weights associated with classes in the form `{class_label: weight}`. If not given, all classes are supposed to have weight one.
- For example: If the frequency of class 0 is 80% and the frequency of class 1 is 20% in the data, then class 0 will become the dominant class and the model will become biased toward the dominant classes. In this case, we can pass a dictionary `{0:0.2,1:0.8}` to the model to specify the weight of each class and the random forest will give more weightage to class 1.
- `bootstrap`: Whether bootstrap samples are used when building trees. If False, the entire dataset is used to build each tree, default=True.
- `max_samples`: If bootstrap is True, then the number of samples to draw from X to train each base estimator. If None (default), then draw N samples, where N is the number of observations in the train data.
- `oob_score`: Whether to use out-of-bag samples to estimate the generalization accuracy, default=False.
- Note: A lot of hyperparameters of Decision Trees are also available to tune Random Forest like `max_depth`, `min_sample_split` etc.

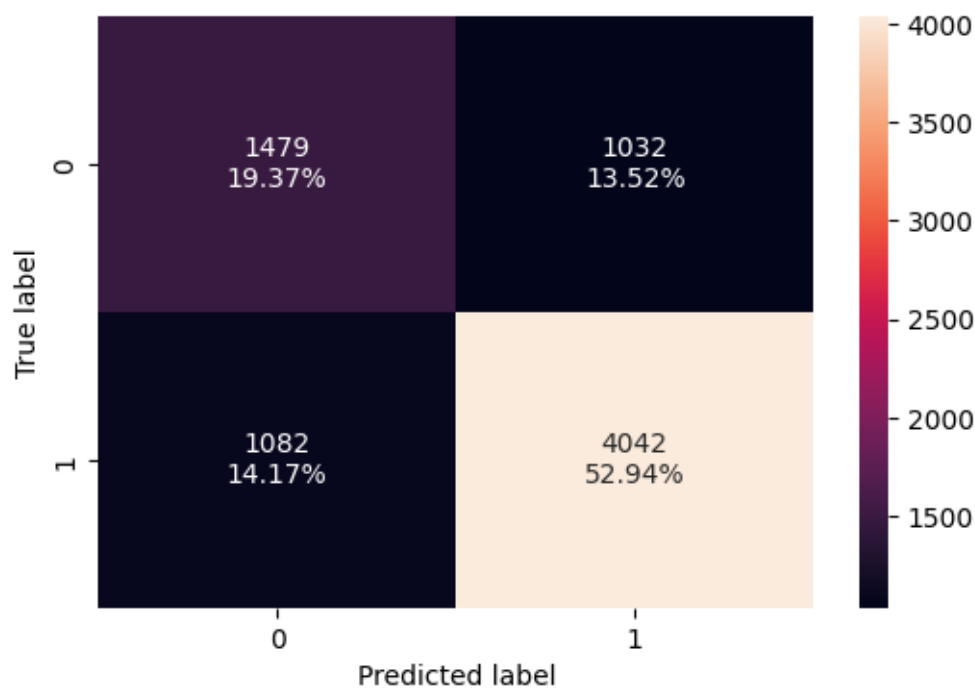


- The Random Forest after tuning performs equally well on the training and testing datasets, indicating that the model is not overfit.
- Additionally, this model performs as well as two out of the three boosting models and the tuned Decision Tree on the test data.
- Both of these indicate this tuned model is a significant improvements over the base model.

Tuning AdaBoost Classifier

Hyperparameters include:

- `base_estimator`: The base estimator from which the boosted ensemble is built. By default the base estimator is a decision tree with `max_depth=1`
- `n_estimators`: The maximum number of estimators at which boosting is terminated. Default value is 50.
- `learning_rate`: Learning rate shrinks the contribution of each classifier by `learning_rate`. There is a trade-off between `learning_rate` and `n_estimators`.

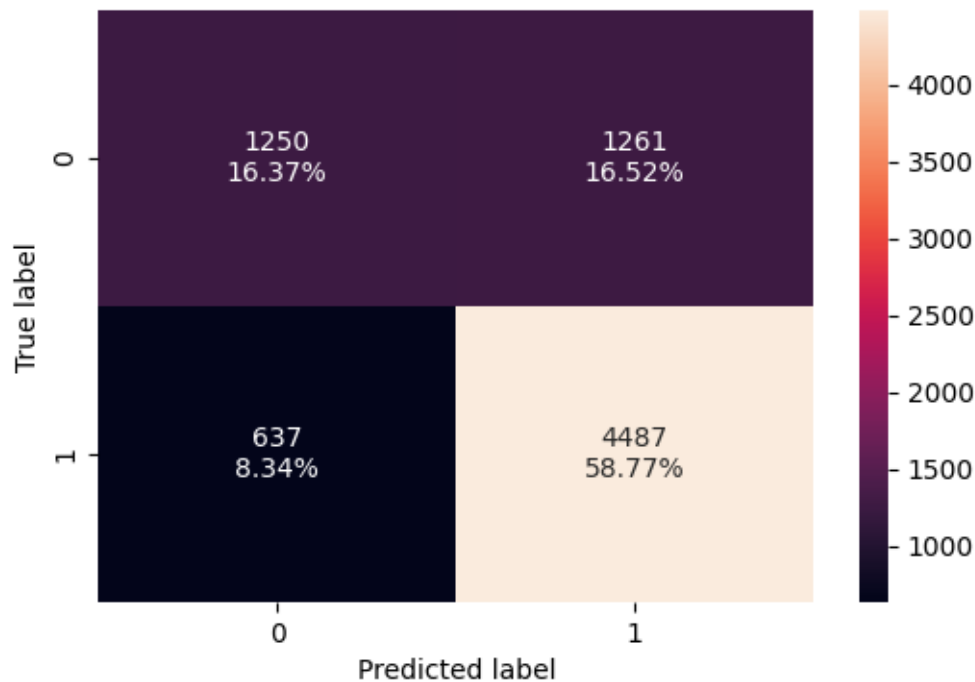


- The AdaBoost classifier after tuning performs equally well on the training and testing datasets, indicating that the model is not overfit.
- However, this model performs worse than the base model on both the train and test data.

Tuning GradientBoost Classifier

Most of the hyperparameters available match those of a random forest classifier.

- `init`: An estimator object that is used to compute the initial predictions. If 'zero', the initial raw predictions are set to zero. By default, a `DummyEstimator` predicting the classes priors is used.
- There is no `class_weights` parameter in gradient boosting.

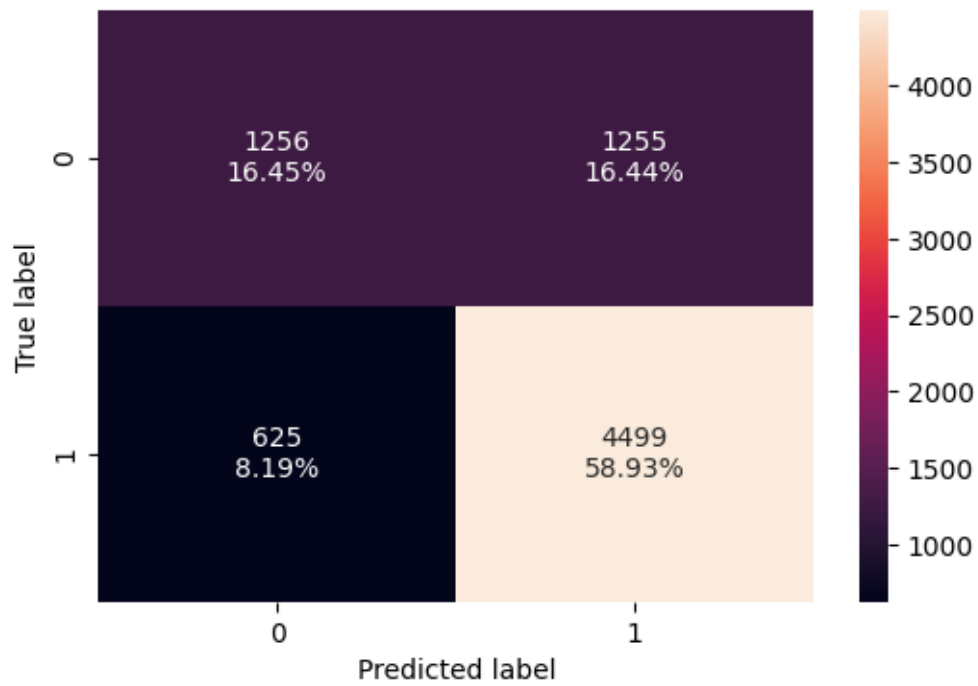


- The Gradient Boosting classifier after tuning performs equally well on the training and testing datasets, indicating that the model is not overfit.
- However, this model performs slightly worse than the base model on both the train and test data.

Tuning XGB Classifier

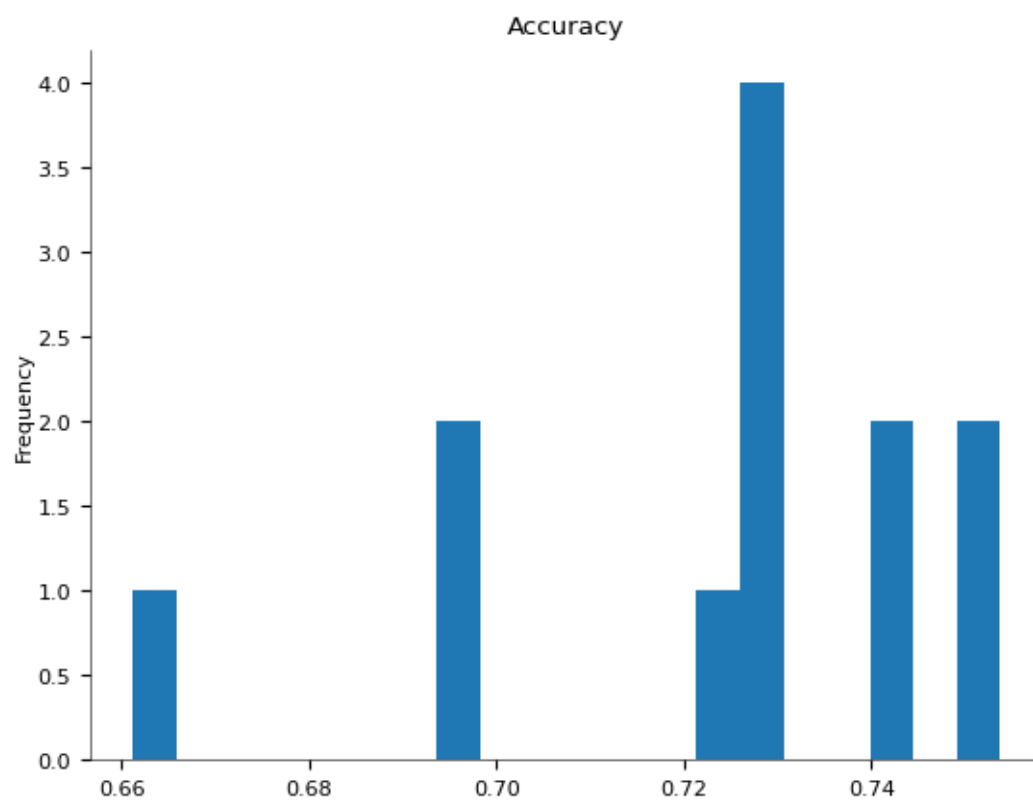
Important parameters include:

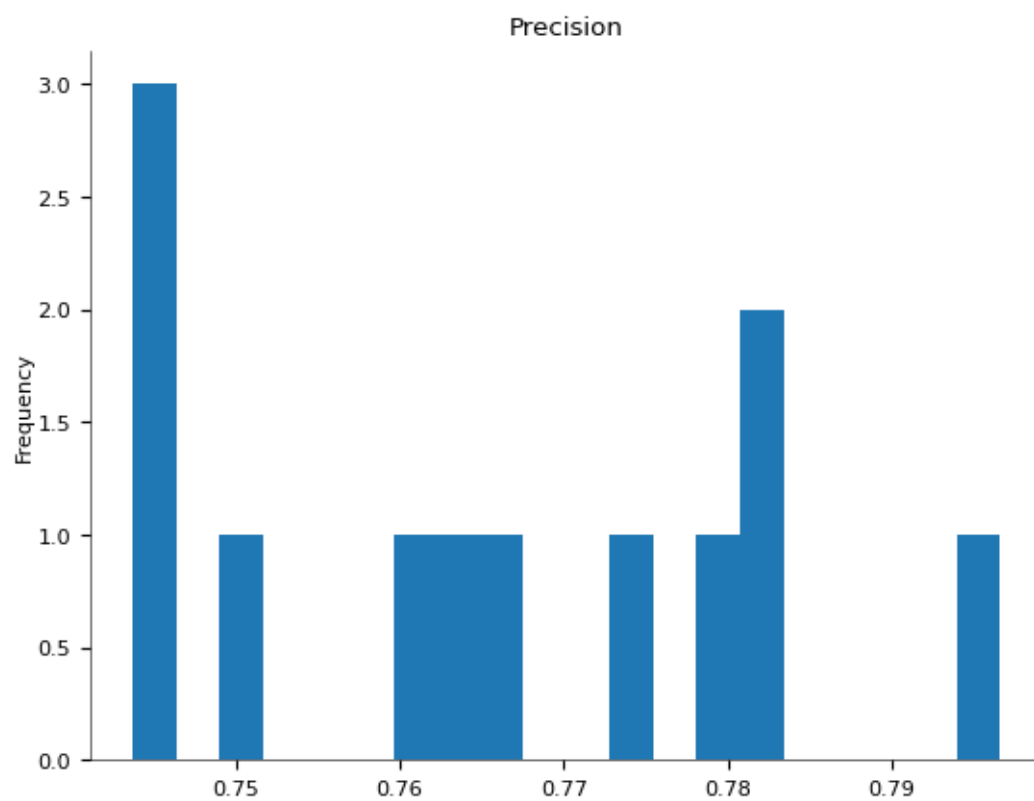
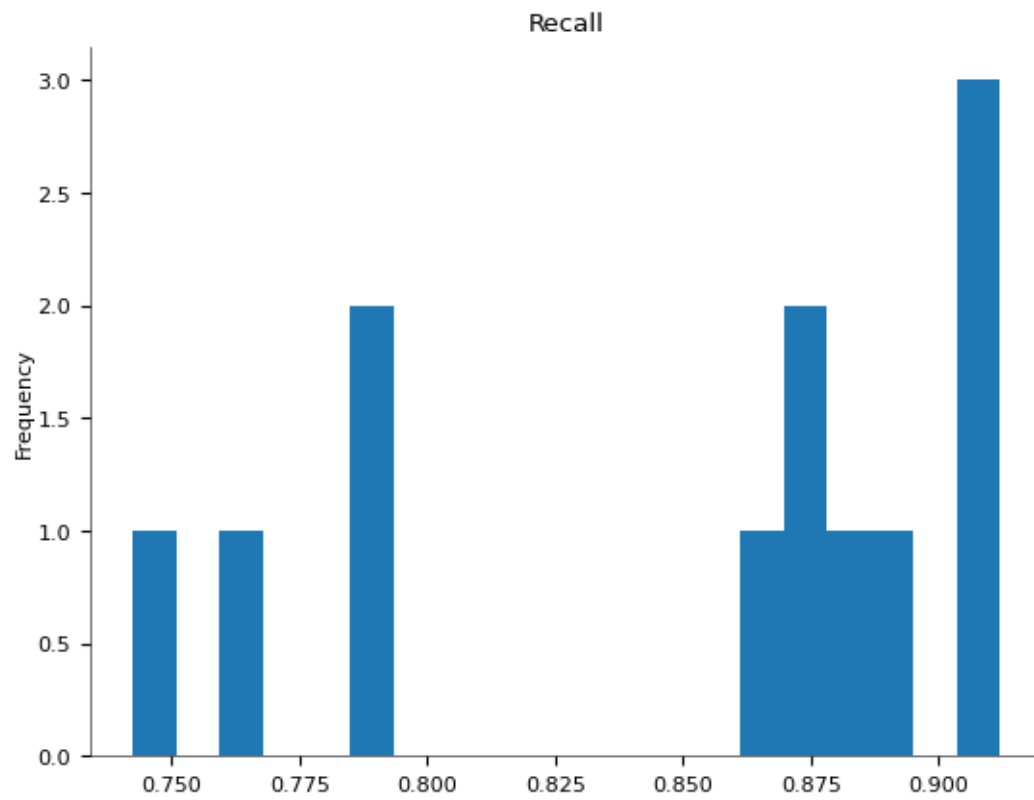
- `scale_pos_weight`: Control the balance of positive and negative weights, useful for unbalanced classes. It has range from 0 to ∞ .
- `subsample`: Corresponds to the fraction of observations (the rows) to subsample at each step. By default it is set to 1 meaning that we use all rows.
- `colsample_bytree`: Corresponds to the fraction of features (the columns) to use.
- `colsample_bylevel`: The subsample ratio of columns for each level. Columns are subsampled from the set of columns chosen for the current tree.
- `colsample_bynode`: The subsample ratio of columns for each node (split). Columns are subsampled from the set of columns chosen for the current level.
- `max_depth`: is the maximum number of nodes allowed from the root to the farthest leaf of a tree.
- `learning_rate/eta`: Makes the model more robust by shrinking the weights on each step.
- `gamma`: A node is split only when the resulting split gives a positive reduction in the loss function. Gamma specifies the minimum loss reduction required to make a split.

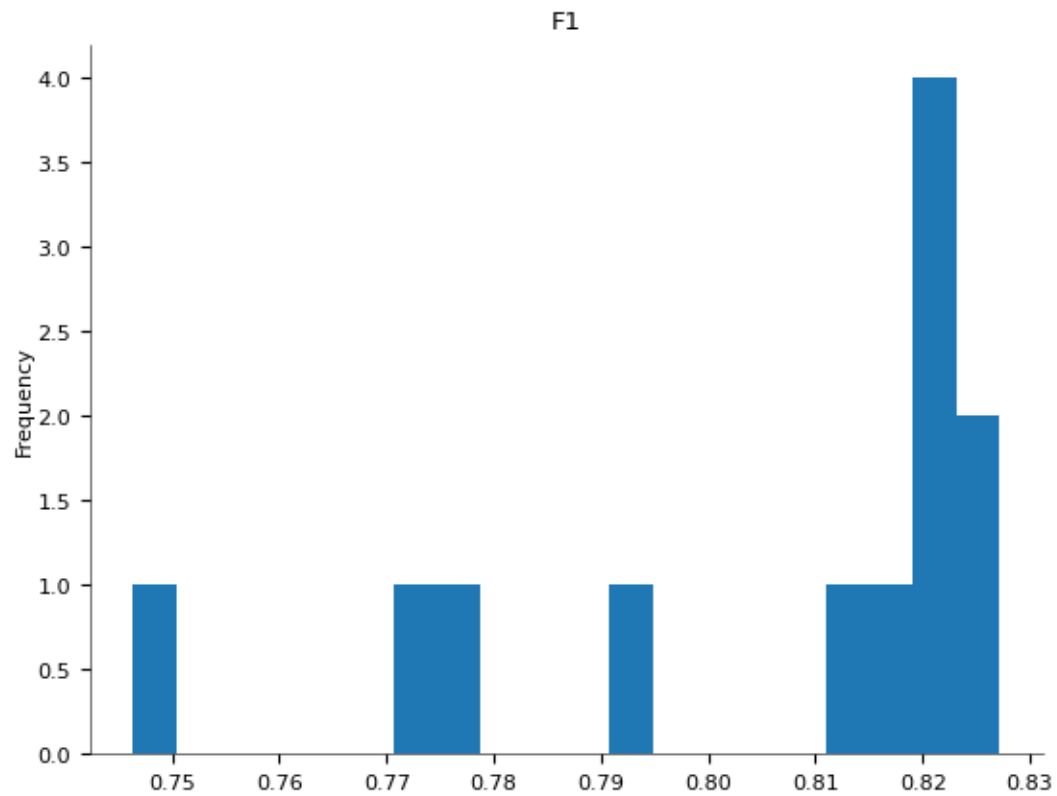


- The XGBoost classifier after tuning is less overfit than the base model, but performs equally well on the test data.
- This indicates a minor improvement over the base model.

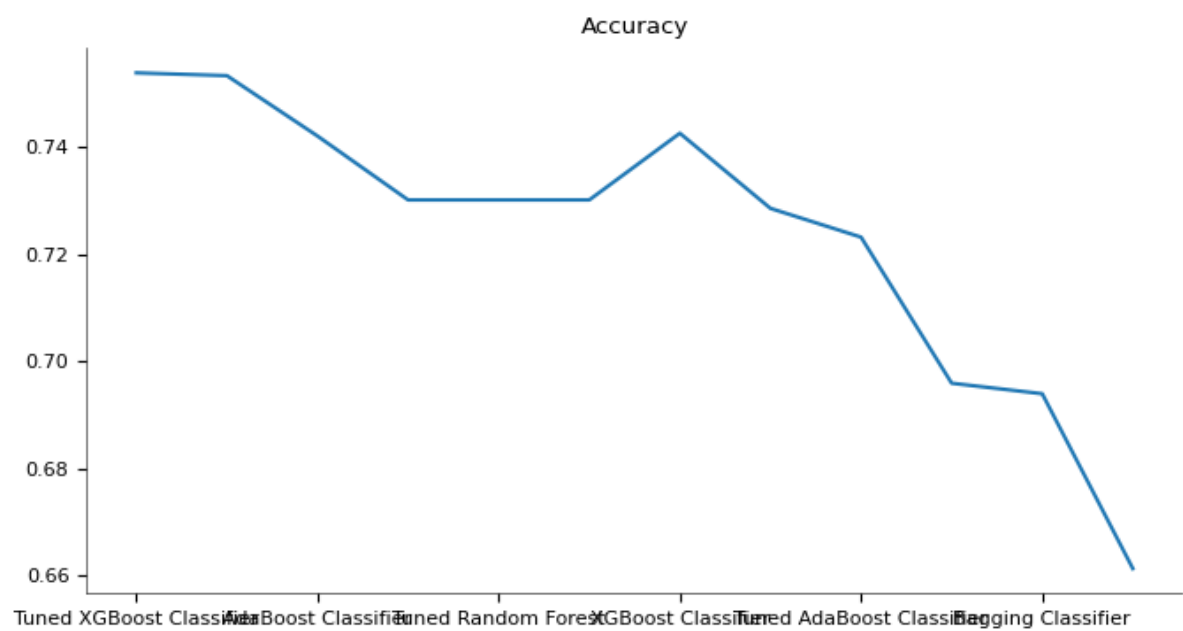
Tuned models comparison

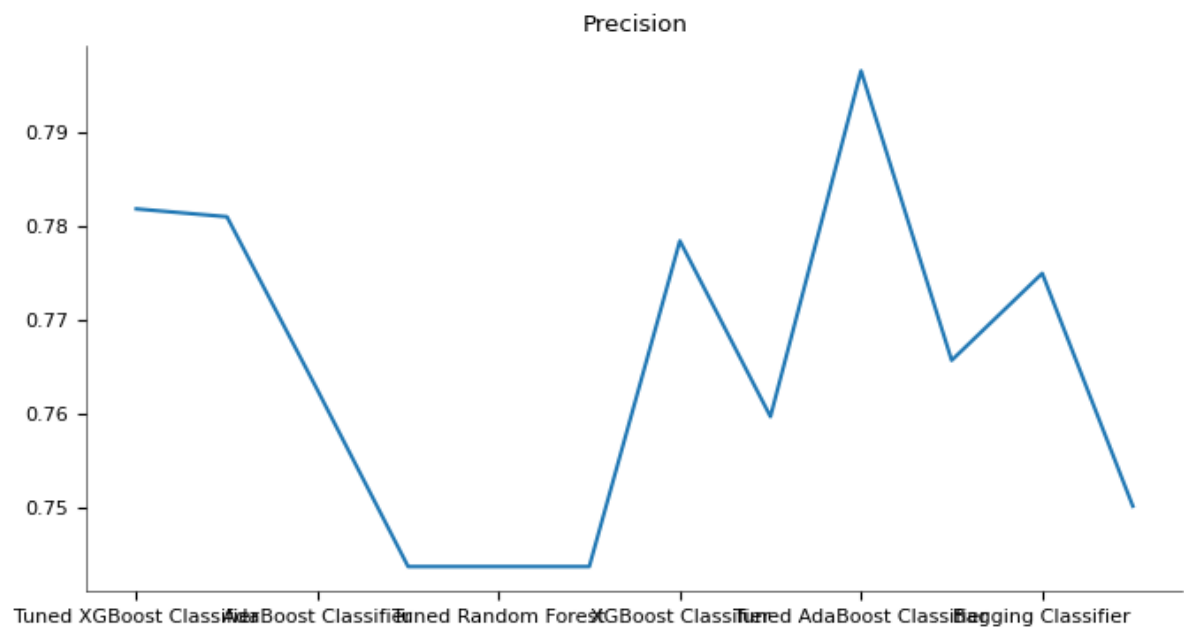
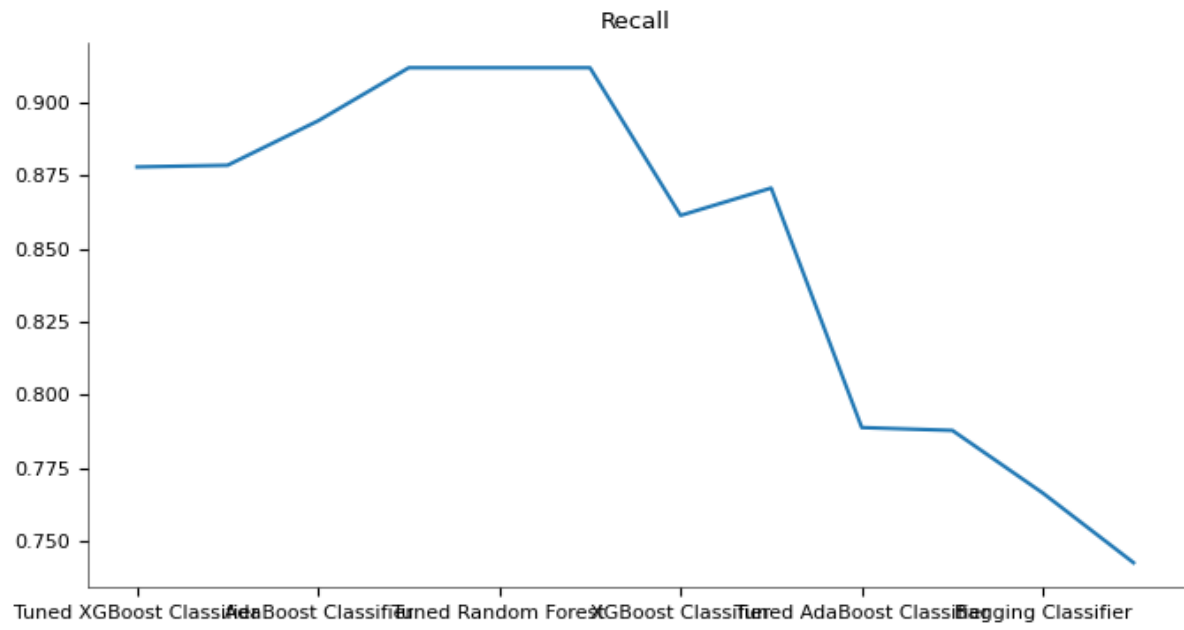


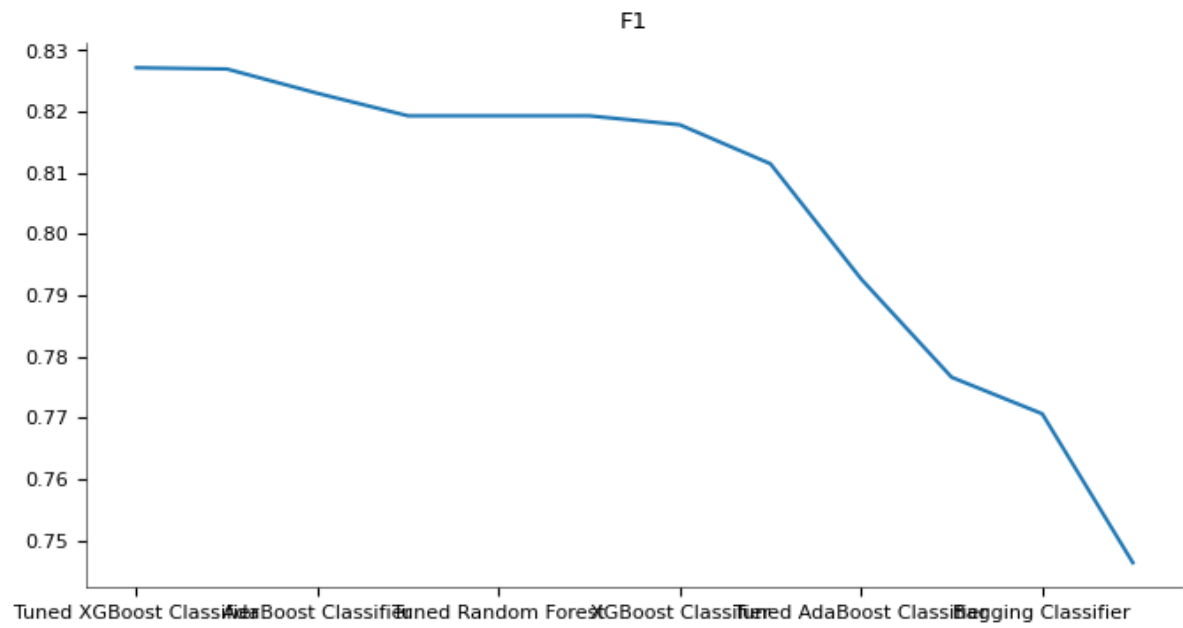




Values





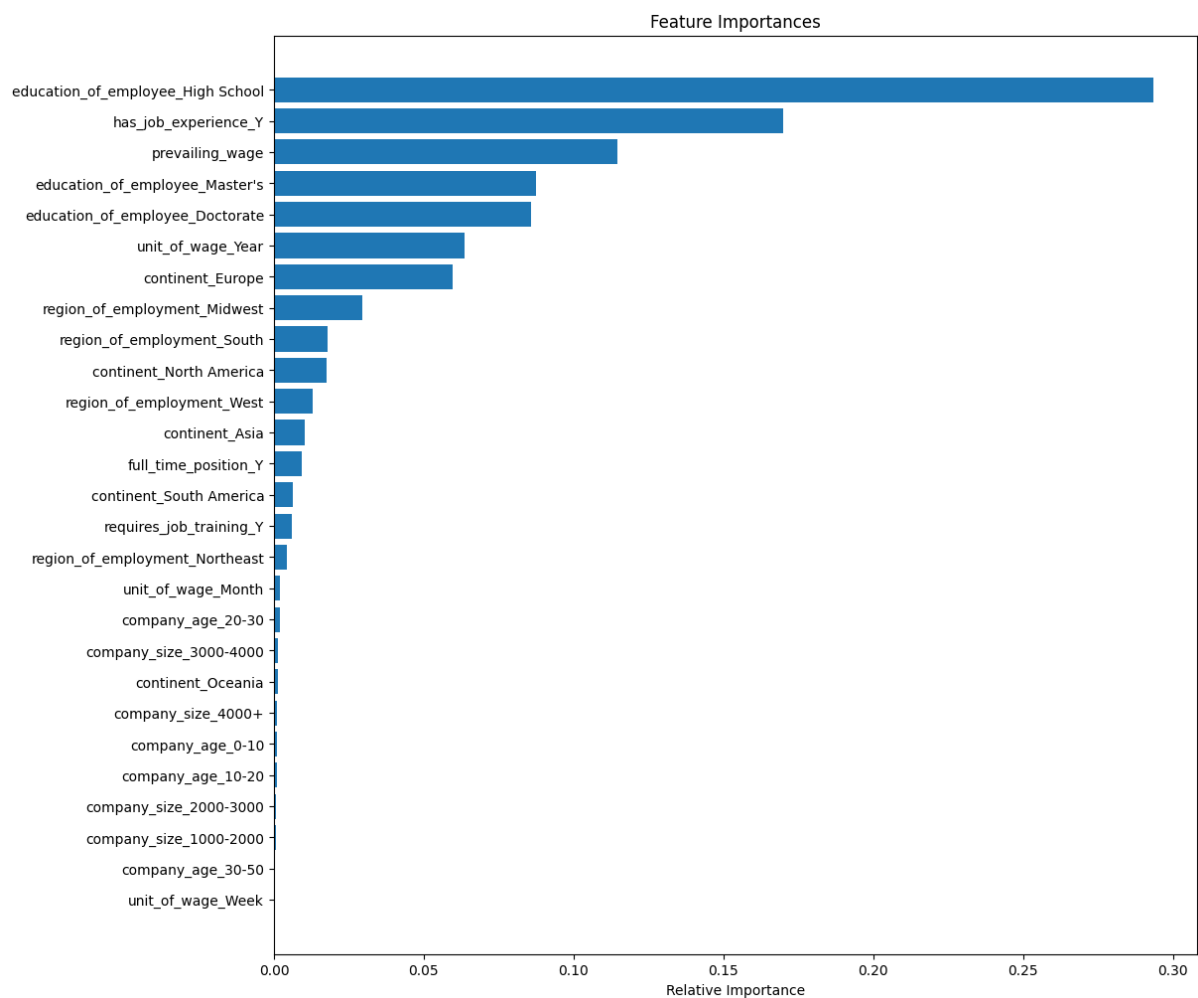


The best performing model on the testing data is the Tuned XGBoost Classifier.

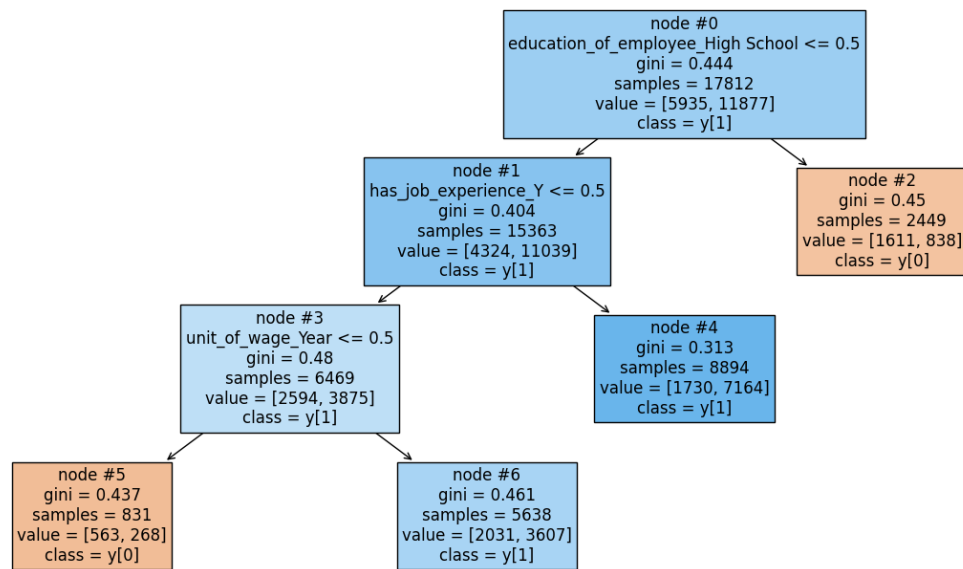
Observations

- As expected the Decision Tree model performs the best on the training dataset, since the model will significantly overfit this data unless pruned otherwise.
- The base Gradient Boosting classifier performs the best on the testing dataset, with an 0.83 F1 score.
- Interestingly, **six** models tied for the second-best performance, with an F1 score of 0.82.

Feature importance of: Gradient Boosting Classifier



Visualization of Decision Tree (tuned)



Key Insights

1. Education Level Matters

- Applicants with **only a high school diploma** are **more likely to be denied**.
- Those with a **Master's degree or higher** have a **high probability of approval**.

2. Prior Job Experience is Critical

- **Applicants without work experience** are more likely to be denied.
- **Experienced candidates** have a **higher chance of approval**.

3. Higher Prevailing Wage Increases Approval Chances

- Jobs with **higher wages** tend to have **higher visa certification rates**.
- **Hourly wage positions** show more variability in approval rates compared to annual salary positions.

4. Company Size & Establishment Year Impact Decisions

- **Larger employers** (with more employees) tend to have **higher approval rates**.
- **Startups and newer companies** face slightly higher rejection rates.

5. Potential Bias in the System

- **Entry-level, less-skilled, and hourly-wage applicants** appear to face **higher rejection rates**, suggesting a possible systemic bias.
- This needs further evaluation to ensure fairness in the visa approval process.

Recommendations

To optimize visa processing and **prioritize applications efficiently**, OFLC should:

1. **Prioritize High-Education Applicants**

- Sort applications by education level and **review Master's and Doctorate holders first**.

2. **Focus on Applicants with Job Experience**

- **Prioritize reviewing applications with prior work experience**, as they have a higher likelihood of approval.

3. **Sort by Wage Structure**

- Divide applications into **hourly-wage and annual-salary jobs**.
- Within each group, **review higher-wage applications first**.

4. **Use Machine Learning to Streamline Decision-Making**

- Implement a **Gradient Boosting model** for efficient, high-accuracy predictions.
- Consider a **Decision Tree model** for greater transparency in decision-making.

5. **Address Bias in Entry-Level and Hourly Wage Jobs**

- Further investigate why **less-skilled and hourly-wage applicants face higher rejection rates**.
- Ensure that the model does not **unfairly disadvantage specific applicant groups**.

By implementing these strategies, OFLC can **reduce processing time, improve efficiency, and ensure a fair, data-driven visa approval process**.

Conclusions

The increasing number of visa applications has made it difficult for the **Office of Foreign Labor Certification (OFLC)** to efficiently process and evaluate applications manually. By leveraging **machine learning models**, we can streamline the visa approval process and improve decision-making efficiency.

The analysis reveals that visa approvals are strongly influenced by **education level, job experience, and prevailing wage**. These three factors can be used to prioritize applications and optimize the review process.

While the **Gradient Boosting Classifier** delivered the best performance (F1-score = 0.83), a **Decision Tree Model** provides a simpler alternative for interpretability. If transparency in decision-making is a priority, the **Decision Tree Model** should be considered despite its slightly lower performance.

Additionally, the model suggests **potential bias** against **entry-level, less-skilled, and hourly-wage applicants**, which requires further investigation to ensure a fair and equitable visa approval system.

THANK YOU

HIYA SHAH